Stellar Cyber 6.6.0s is now live! Discover all the updates in the release notes!

Configuring Modular Sensor Profiles 6.4.0

Learn more at Stellar Cyber Academy.

The following link takes you to a course on the Stellar Cyber Academy technical training portal where you can learn more about this topic by watching the suggested lessons.

M03 Modular Sensor Concepts and Configuration (08h:56m)

Modular Sensor Profiles (16m:00s)

Set up and customize Modular Sensor profiles to configure various features. Assign profiles to sensors and understand best practices for using profiles to control log forwarding, network traffic analysis, and IDS functionality.

Modular Sensor Specific Configuration (24m:00s)

Configure individual Modular Sensors within the Stellar Cyber Platform by setting key parameters such as tenant name, hostname, and timezone. Follow along with a practical demonstration on creating, cloning, and assigning sensor profiles, then explore how to edit profile settings and apply them to specific sensors. Learn to validate sensor status and profile assignments for efficient system management.

Log Forwarder and Log Filter Configuration (15m:00s)

Configure log forwarders to ingest data from third-party systems and optimize ingestion with log filters. Learn how Modular Sensors normalize logs into Interflow records and forward them to the Stellar Cyber Platform for analysis, enabling efficient threat detection.

Log Forwarder Example (38m:00s)

Learn how to configure a Modular Sensor to forward logs from a pfSense firewall to the Stellar Cyber Platform. Follow step-by-step guidance for setting up log parser ports, validating connections, and normalizing logs into Interflow records stored in the data lake. Watch a demonstration of the full configuration process, and explore techniques for troubleshooting common issues to ensure smooth data ingestion and support effective security analysis and threat detection.

Intrusion Detection System (IDS) Configuration (11m:00s)

Learn how to configure IDS on Modular Sensors and explore how the Stellar Cyber Platform processes the resulting event data. Discover how IDS analyzes network traffic using commercial rules, generates Interflow records, and forwards metadata to the data lake for further analysis. Explore how machine learning enhances raw IDS output to produce actionable alerts, enrich findings with threat intelligence, and correlate events across data sources for comprehensive detection and response.

Malware Sandbox (10m:00s)

Learn to configure the malware sandbox module on Modular Sensors and understand how sandboxing enhances threat detection by analyzing suspicious files in an isolated environment. Explore how the Stellar Cyber Platform ingests sandbox data into the data lake alongside Interflow records to enrich threat intelligence and support machine learning-driven analysis. Discover how to navigate the malware sandbox interface to review file metadata, sandbox actions, and malware categorizations, and follow a demonstration using toolbar filters and the search tool to detect patterns in file analysis results. See how to interpret sandbox findings and integrate them into threat investigations and Case Management workflows for a more complete response to potential malware activity.

Modular Sensor Aggregator (21m:00s)

Learn how to configure the Modular Sensor Aggregator to consolidate data streams from multiple sensors into the Stellar Cyber Platform. Follow a demonstration of setting up aggregation settings and validating data flow, and explore how aggregation optimizes throughput for efficient processing and threat detection.

Modular Sensor Tenable Nessus Scanner (30m:00s)

Learn how to integrate Tenable Nessus with Modular Sensors to enhance vulnerability scanning capabilities within the Stellar Cyber Platform. Explore how scan data is normalized into Interflow records and ingested into the data lake for threat correlation and analysis. Follow a demonstration of configuring the Tenable Nessus integration, validating data ingestion, and verifying that vulnerability findings appear correctly across the platform.

Modular Sensor Buffering (08m:00s)

Understand the buffering capabilities of Modular Sensors, designed to temporarily store data during connectivity issues. Learn how buffered Interflow records are trickled back to the Stellar Cyber Platform once the connection is restored.

The first time you access a link on the portal during a session, you must log in to access content.

A modular sensor profile allows you to easily add the features you like to your profiles. This helps simplify your profiles and lets you manage the VM requirements for the sensors based on the modular features they use. As we add new modular features, you can choose to add them to your sensor profiles.

Modular sensor profiles can only be applied to modular sensors. You can:

- Add a modular sensor profile

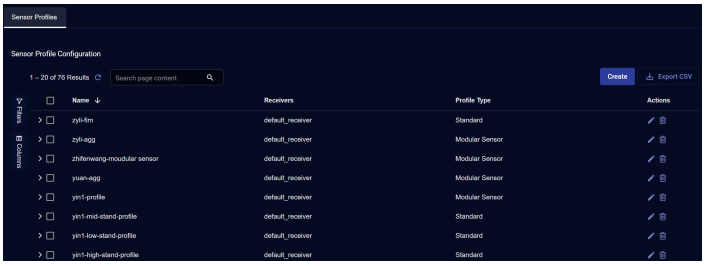
- Customize the profile features

Keep in mind that VM resou requirements increase as you add more features to the Modular Sensor Profile. Refer to **Virtual Modular Sensor Provisioning and Performance** for details on the resources required to run different combinations of features in a Modular Sensor Profile, as well as how to use the **show module** and **show module request** CLI commands to compare provisioned resources against those required to run specific feature combinations. Stellar Cyber only enables a Modular Sensor Profile on a sensor if the host VM's resources can support it.

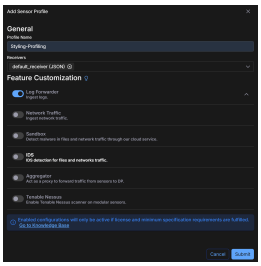
Adding a Modular Sensor Profile

Whether you add or edit a modular sensor profile you have the same options, except you cannot edit the profile name. Use the following procedure to add a modular sensor profile:

1. Click **System | DATA SOURCE MANAGEMENT | Sensors | Sensor Profiles**. The **Sensor Profile Configuration** page appears, with the **Sensor Profiles** tab displayed by default.



2. Click **Create** and select **Add Modular Sensor Profile**. The **ADD SENSOR PROFILE** screen appears.



3. Enter the **Profile Name**. We recommend that you establish a naming convention so you can easily understand the intent of each profile by looking at the name. This field can only contain alphanumeric characters, underscores, spaces, and dashes.
4. Choose at least one **Receiver**. Each sensor profile must have at least one receiver, which is the destination of the data it collects.

   See the Receiver configuration page for more information on creating and maintaining receivers. Keep in mind that Modular Sensors only use **JSON** receivers. They do not use **Packet** receivers.
5. Customize the settings for the profile. The available settings are explained below.
6. Click **Submit**. The profile is active immediately .

Keep in mind the following:

- **Log Forwarder** is always enabled in Modular Sensor Profiles.

- The **Sandbox** and **IDS** (Intrusion Detection System) features can only be enabled if **Network Traffic** is enabled and if their corresponding licenses are enabled.

  To enable their licenses, select **System | DATA SOURCE MANAGEMENT | Sensors | Sensors | Manage Authorizations**, select **APT License** (Advanced Persistent Threat) for Malware Sandbox and **IDS License** license for IDS, and then **Submit**.

Customizing Modular Sensor Profile Features

You can customize these modular sensor profile features:

- Log forwarder
- Network traffic
- Sandbox
- IDS
- Aggregator
- Tenable Nessus

Log Forwarder

> See the top of this page for recommended lessons about this section at Stellar Cyber Academy.

Modular sensor profiles include a Log Forwarder by default.

- Prior to the 6.4.0 release, the **Log Forwarder** option was always on and could not be disabled.
- Starting with 6.4.0, the **Log Forwarder** option can be toggled off and on. This lets you conserve sensor memory by disabling log forwarding on Modular Sensors that do not require it (for example, on a sensor that is used primarily for security features, such as **IDS** and **Malware Sandbox**). 6.4.0

You can configure the following options for the log forwarder:

- Workers
- Data filtering
- Forward to external server
- Compression
- HTTP JSON parser
- Multi-Tenant Log Ingestion
- Raw Log Capture
- Buffering

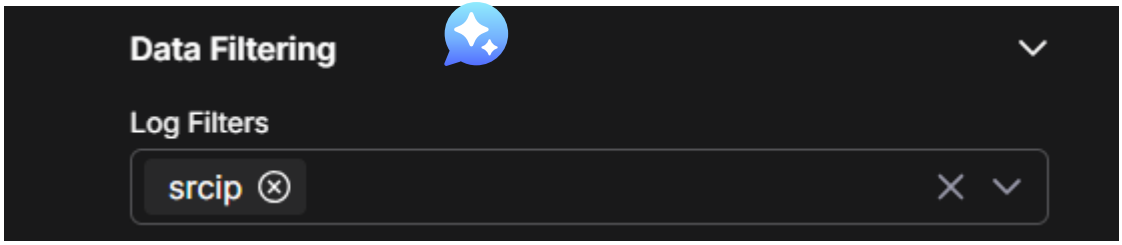
Log Forwarder Workers

Configure the record batch size for log forwarder workers:

- **Batch Size** – Set the record batch size. The default is 100 records. The range is 1-10000.

  The sensor batches log records to make transmission more efficient. If your network has high throughput, increase the **Batch Size** to increase that efficiency.

Data Filtering



Add log filters to filter traffic out before sending it to the DP. You can create log filters at **System | DATA SOURCE MANAGEMENT | Data Filters | Log Filters**.

Stellar Cyber recommends configuring no more than 50 Log Forwarder filters in this field for one sensor profile. Exceeding this recommended maximum can affect sensor performance.

Forward to External Server

You can enable this option to send unparsed logs to an external server, pre-normalization. Logs are still also sent to the DP when this feature is enabled.

Configure external log forwarding as follows:

1. Click **Forward to External Server**.
2. Enter the IP address or domain name of the external server to which you want to forward unparsed logs.
3. Enter the port.
4. Uncheck **Send Metadata** if you don't want the sensor to add metadata to the logs forwarded to your external server. If you leave it checked, the sensor adds the log source IP address, ingestion port number, and log source type to the original log in JSON format. This adds roughly 100 bytes to each log.

Logs containing non-printing characters, such as Netflow and IPFIX, cannot be forwarded to an external server.

Starting with the 5.1.1 release, logs are truncated to 2048 characters, including metadata, if you have enabled it (metadata adds roughly 100 bytes to each log).
For previous releases, logs are truncated to 1024 characters, per RFC 3164.

Compression

Enable **Compression** for the sensor to compress the parsed logs before sending to the DP. This uses the CPU on the sensor to compress the logs, and the CPU on the DP to decompress. Enable compression to save bandwidth at the expense of CPU on the sensor and DP.

HTTP JSON Parser

Enable the **HTTP JSON Parser** if you are sending logs via HTTP.

Multi-Tenant Log Ingestion

Enable Multi-Tenant Log Ingestion to allow CEF (Infocyte), Cylance, and selected Stellar JSON parsers to receive logs from multiple tenants. In addition to this setting, the incoming data

file must follow specific guidelines described in: **Single Sensor Multi-Tenant Log Ingestion** . Supported parsers are also described in this topic.

Raw Log Capture

Raw Log Capture can be enabled if you want the sensor to store both raw and processed logs on the DP for built-in parsers. This can be useful in troubleshooting situations when you may want to compare a processed log to its corresponding raw log. Note that the feature only applies to built-in parsers. Custom parsers have their own settings within the parsers themselves.

You can search for raw logs by looking for the `parser_raw_msg` field in InterFlow records.

Keep in mind that when this feature is enabled, built-in parsers store both raw and processed logs, requiring more storage resources on the DP.

Buffering

See the top of this page for recommended lessons about this section at Stellar Cyber Academy.

Enable this to enable buffering on the data sensors. If you enable buffering, the data sensor buffers the logs sent to Stellar Cyber. If the logs are received successfully, the sensor deletes the buffered logs. If the buffer is full, the sensor stops buffering logs. The Buffer Size parameter lets you specify the maximum size for all logs buffered:

- **Buffer Size** – The total logs buffered, in MB

The buffer size is specified in **MB**, even though the Sensor Profile user interface may incorrectly state **KB** for the units.

When you disable buffering, any data in the buffer is immediately deleted. Ensure that the sensor has a stable connection to the DP for long enough to transmit all data in the buffer.

Network Traffic

Enable the **Network Traffic** option to ingest network traffic. When this option is enabled, you can also enable the **Sandbox** and **IDS** features. You can configure the following options:

- Pre-DPI Filter
- Application identification
- Application session
- Application metadata
- Process correlation
- Packet deduplication
- Stream slicing
- Handshake failure
- Flood attack
- Buffering

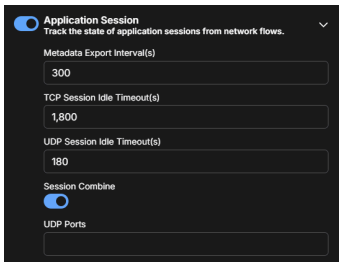
Pre-DPI Filter

Enable this option to optimize sensor performance by removing unnecessary traffic before doing Deep Packet Inspection (DPI). F configuration details, see Pre-DPI Filters .

Application Identification

Enable this option for the sensor to identify applications associated with events.

Application Session

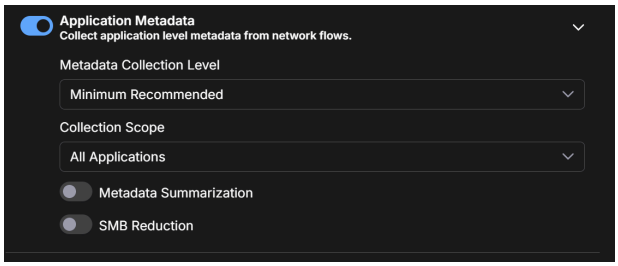


Enable this option for the sensor to identify the length of individual network sessions. The parameters specify how often to report the session information and the amount of idle time that can pass before concluding that a session has ended.

If you disable **Application Session**, network and agent sensors stop sending data to the DP but data sensors continue to process traffic.

If you enable **Session Combine**, UDP sessions from the specified port to the same destination are combined into a single session.

Application Metadata



Enable this to control how application information is collected and which applications are included.

You can disable the collection of application metadata to increase sensor performance and save storage space, but that limits collected data to:

- Basic session information
- Packet and byte counts
- IP address
- Port

You can set the **Metadata Collection Level** to **Limited Visibility**, **Minimum Recommended**, **Standard**, or **Maximum**. This controls how many of the information fields are collected.

Click to see the fields collected by **Limited Visibility**:

- 0zz0 login
- 0zz0 password
- activesync login

- aim_express contact_login
- aim_express login
- aims login
- archive login
- archive password
- badongo login
- badongo password
- badoo contact_login
- badoo login
- badoo password
- bebo login
- bebo password
- dhcp agent
- dhcp chaddr
- dhcp ciaddr
- dhcp client_fqdn
- dhcp domain_name
- dhcp dns_server
- dhcp end
- dhcp end_status
- dhcp giaddr
- dhcp host_name
- dhcp inherit_parent
- dhcp ip_lease_time
- dhcp message_type
- dhcp option
- dhcp option_type
- dhcp option_value_buffer
- dhcp request
- dhcp router
- dhcp siaddr
- dhcp sname
- dhcp subnetmask
- dhcp xid
- dhcp yiaddr
- dimp login
- dimp password
- dns dns_entry
- dns dns_query
- dns end
- dns tunneling
- dns flags
- dns host
- dns host_addr
- dns host_addr6
- dns host_type

- dns message_type
- dns name
- dns query
- dns query_type
- dns reply_code
- dns reverse_addr
- dns reverse_addr6
- dns section_type
- dns transaction_id
- dns ttl
- ebuddy contact_login
- ebuddy login
- ebuddy password
- facebook login
- facebook password
- facebook_mail login
- friendster login
- friendster password
- ftp content_type
- ftp data_port
- ftp filename
- ftp filesize
- ftp index
- ftp loadway
- ftp login
- ftp password
- ftp method
- ftp return_code
- ftp transfer_duration
- gmail_basic login
- gmail_chat login
- gmail_mobile login
- gmail_mobile password
- gmx login
- hi5 login
- hi5 password
- http auth_username
- http auth_password
- http code
- http index
- http method
- http proxy_auth
- http proxy_login
- http proxy_realm
- http referer
- http server

- http uri
- http filename
- http file_type
- http forward_addr
- http forward_addr6
- http host
- http mime_type
- http part_content_type
- http user_agent
- ica login_info
- ica login_info_utf16
- icmp code
- icmp message
- icmp rtt
- icmp type
- kakaotalk login
- krb5 error_code
- krb5 login
- krb5 server
- krb5 message_type
- krb5 service
- laposte_webmail login
- ldap hostname
- ldap name
- linkedin login
- live_groups login
- live_hotmail login
- livemail_mobile login
- mailru login
- maktoob login
- mimp login
- mimp password
- msn contact_login
- msn login
- mysql login
- mysql query
- mysql request
- mysql result_set
- mysql sqlstate_code
- mysql number_columns
- mysql number_rows
- mysql query_id
- netflix login
- netlog login
- netlog password
- nfs filename

- nfs filesize
- nfs type_string
- nfs symlink_name
- nfs uid
- niconico_douga login
- nntp login
- nntp password
- odnoklassniki login
- odnoklassniki password
- oovoo login
- orangemail login
- owa login
- paltalk contact_login
- paltalk login
- paltalk_transfer login
- pap login
- pap password
- perfspot login
- perfspot password
- pop3 password
- postgres authentification_type
- postgres error
- postgres login
- postgres password
- postgres query
- postgres query_id
- postgres sqlstate_code
- postgres number_columns
- postgres number_rows
- qq contact_login
- qq login
- qq_transfer login
- radius login_ipv6_host
- radius login
- rambler login
- rambler password
- rambler_webmail rambler_webmail
- rapidshare login
- rapidshare password
- rdp default_username
- rdp username_ascii
- rdp username_raw
- rlogin client_login
- rlogin server_password
- rsh login
- rsh remote_login

- rsync login
- rsync password
- rtmp page_url
- secondlife login
- skyblog login
- skyblog password
- slsk login
- slsk password
- smb login
- smb user_id
- smb nt_status
- smb command_string
- smpp receiver
- smpp sender
- smtp login
- smtp password
- smtp response_code
- snpp login
- snpp password
- socks5 login
- socks5 password
- spdy server_agent
- spdy user_agent
- squirrelmail login
- squirrelmail password
- ssh version
- ssl common_name
- ssl issuer
- ssl subject_alt_name
- ssl validity_not_after
- ssl version
- ssl validity_not_before
- tagged login
- tagged password
- tango login
- tchatche login
- tchatche password
- tds login
- tds password
- tds sqlstate_code
- tds query
- tds query_id
- tds number_columns
- tds number_rows
- teamspeak login
- teamspeak password

- telnet login
- telnet password
- tftp filename
- tftp filesize
- tftp query
- tftp request_filename
- tns login
- tns password
- tns query
- tns sqlstate_code
- twitter login
- twoo login
- twoo password
- ustream login
- ustream password
- viadeo login
- viadeo password
- vkontakte login
- vkontakte password
- winmx login
- yahoo_groups login
- yandex login
- yandex password
- yandex_webmail login
- ymail2 login
- ymail_classic login
- ymail_mobile_new login
- ymail_mobile_new password
- ymsg_conf login
- ymsg contact_login
- ymsg login
- ymsg_transfer login
- ymsg_video login
- ymsg_webmessenger contact_login
- ymsg_webmessenger login
- youtube login
- youtube password
- zimbra login
- zimbra password
- zimbra_standard login
- zimbra_standard password

Minimum Recommended collects everything in **Limited Visibility**. There is currently no difference between **Minimum Recommended** and **Limited Visibility**.

Standard collects everything in **Minimum Recommended** and:

- amqp class

- amqp type
- amqp method
- amqp replyto
- amqp routing_key
- dhcp circuit_id
- dhcp remote_id
- dhcp remote_id_subtype
- dhcp remote_id_type
- dns web_application_info
- dns ancount
- dns arcount
- dns nscount
- dns opcode
- dns qdcount
- ftp return_content
- ftp offset
- ftp inherit_parent
- ftp data_port_end_offset
- ftp data_port_start_offset
- ftp_data inherit_key
- http cache_control
- http connection_header
- http content_disposition
- http content_encoding
- http content_len
- http cookie
- http date
- http end
- http etag
- http expires
- http image_type
- http is_webdav
- http location
- http ntlm_domain
- http ntlm_user
- http ntlm_workstation
- http part_filename
- http part_header_name
- http part_header_value
- http part_name
- http request
- http request_size
- http server_agent
- http set_cookie
- http smb_client
- http version

- http video_type
- http video_url
- icmp seq
- icmp typeval
- icmp wrong_crc
- krb5 err_cname_name
- krb5 err_sname_name
- krb5 realm
- ldap krb5_enc_data
- ldap krb5_enc_data_type
- ldap krb5_err_cname
- ldap krb5_err_cname_name
- ldap krb5_err_cname_type
- ldap krb5_err_crealm
- ldap krb5_err_data
- ldap krb5_error_code
- ldap krb5_err_realm
- ldap krb5_err_sname
- ldap krb5_err_sname_name
- ldap krb5_err_sname_type
- ldap krb5_err_text
- ldap krb5_kdcoptions
- ldap krb5_message
- ldap krb5_message_type
- ldap krb5_pa_data
- ldap krb5_pa_data_type
- ldap krb5_pa_data_value
- ldap krb5_realm
- ldap krb5_server
- ldap krb5_service
- ldap krb5_ticket
- ldap krb5_ticket_enc_part
- ldap krb5_ticket_name
- ldap krb5_ticket_name_type
- mysql end
- mysql bind_variable
- mysql variable_id
- mysql variable_type
- mysql variable_value
- nfs current_state
- nfs gid
- nfs mode
- nfs procedure
- nfs status
- nfs type
- nfs version

- pop3 attach_filename
- pop3 file_type
- pop3 login_server
- pop3 login
- pop3 received_by_name
- pop3 received_date
- pop3 receiver
- pop3 received_from_name
- pop3 receiver_email
- pop3 sender
- pop3 sender_email
- pop3 subject
- postgres variable_format
- postgres variable_id
- postgres variable_type
- postgres variable_value
- postgres base
- postgres proto_version
- postgres server_version
- rpc message_type
- rpc procedure
- rpc program
- rpc xid
- sip authorization_username
- smb filename
- smb filesize
- smb file_type
- smb directory
- smb domain
- smb file_id
- smb host
- smb krb5_error_code
- smb krb5_err_text
- smb krb5_ticket
- smb loadway
- smb native_os
- smb path
- smtp attach_filename
- smtp attach_size
- smtp file_type
- smtp received_by_name
- smtp received_date
- smtp received_from_name
- smtp receiver
- smtp receiver_email
- smtp sender

- smtp sender_email
- smtp server
- smtp subject
- snmp method
- snmp name
- snmp oid
- snmp value_raw
- snmp version
- spdy location
- spdy mime_type
- ssl certif_sha1
- ssl handshake_type
- ssl organization_name
- ssl parent_common_name
- ssl protocol_version
- ssl server_name
- syslog code
- tds bind_variable
- tds variable_id
- tds variable_type
- tds variable_value
- telnet term_type
- tftp mode
- tns client_hostname
- tns server_hostname
- tns bind_variable
- tns variable_id
- tns variable_type
- tns variable_value

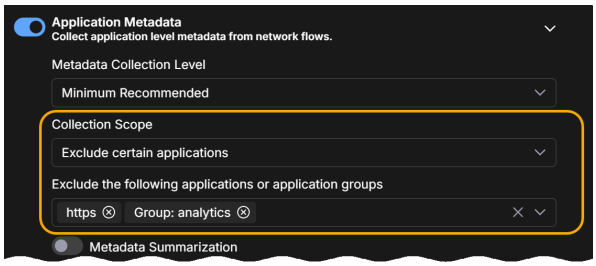
Maximum collects everything in **Standard** and:

- amqp correlation_id
- amqp exchange_type
- amqp revision
- amqp server_major_version
- amqp server_minor_version
- ica application
- ica application_utf16
- ica priority
- ica service
- mysql base
- pop3 attach_disposition
- pop3 attach_transfer_encoding
- pop3 date
- pop3 received_by_ip
- pop3 received_from_ip
- pop3 received_server_agent

- pop3 received_with
- pop3 receiver_alias
- pop3 receiver_type
- pop3 sender_alias
- pop3 user_agent
- rpc program_version
- rpc state
- rpc version
- sip inherit_key
- sip inherit_parent
- sip method
- sip reply_code
- sip request_call_id
- sip user_agent
- smb attributes
- smb ext_attributes
- smb file_attributes
- smb file_chunk_data_offset
- smb file_chunk_len
- smb krb5_enc_data
- smb krb5_enc_data_type
- smb krb5_err_cname
- smb krb5_err_cname_name
- smb krb5_err_cname_type
- smb krb5_err_crealm
- smb krb5_err_data
- smb krb5_err_realm
- smb krb5_err_sname
- smb krb5_err_sname_name
- smb krb5_err_sname_type
- smb krb5_kdcoptions
- smb krb5_message
- smb krb5_message_type
- smb krb5_pa_data
- smb krb5_pa_data_type
- smb krb5_pa_data_value
- smb krb5_realm
- smb krb5_server
- smb krb5_service
- smb krb5_ticket_enc_part
- smb krb5_ticket_name
- smb krb5_ticket_name_type
- smb search_attributes
- smb search_pattern
- smb security_blob
- smb service

- smb service_data
- smb session_key
- smb share_access
- smtp attach_disposition
- smtp attach_transfer_encoding
- smtp content_language
- smtp content_transfer_encoding
- smtp date
- smtp received_by_ip
- smtp received_from_ip
- smtp received_server_agent
- smtp received_with
- smtp receiver_alias
- smtp receiver_type
- smtp replyto
- smtp sender_alias
- smtp server_agent
- smtp start_time
- smtp stop_time
- smtp user_agent
- tns base
- tns client_os
- tns client_program_name
- tns client_program_path
- tns response_size
- tns server_os
- tns server_port
- tns version

Collection Scope allows you to filter specific metadata that is sent to the Data Processor.



- Select **All applications** for no traffic filtering
- Select **Exclude certain applications** to collect metadata for all content except the applications and application groups you specify in the selector box. .
- Select **Only collect certain applications** to exclude all metadata except that for the applications and application groups you specify in the selector box.

When you select an option other than **All applications**, the user interface updates for you to then **Exclude** or **Include the following applications or application groups.** Begin typing in the selector box to display a list of configured **Application Groups** and **Applications**. **Applications Groups** are listed first. Scroll through the list to get to the set of **Applications**. When you select

an Application Group, the label for it is has a prefix (**Group:**) to distinguish it from Applications. In the image above, there is one application and one group.

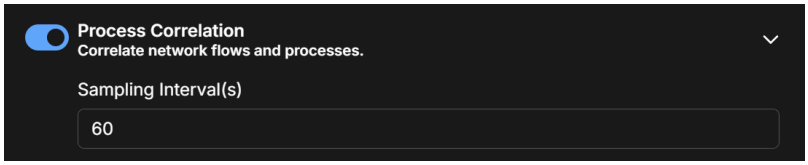
In previous releases, Application Groups were referred to as Tags.

Metadata Summarization allows you to reduce traffic to the DP. If you enable this, the sensor groups similar metadata from chatty applications into a single JSON file before sending to the DP.

Finally, you can enable **SMB Reduction** to reduce the amount of metadata collected for SMB commands. This is especially helpful in increasing your compression ratio if your network has a great deal of SMB traffic. If you enable this, metadata is only collected for the following:

- session setup
- logoff
- read
- write

Process Correlation

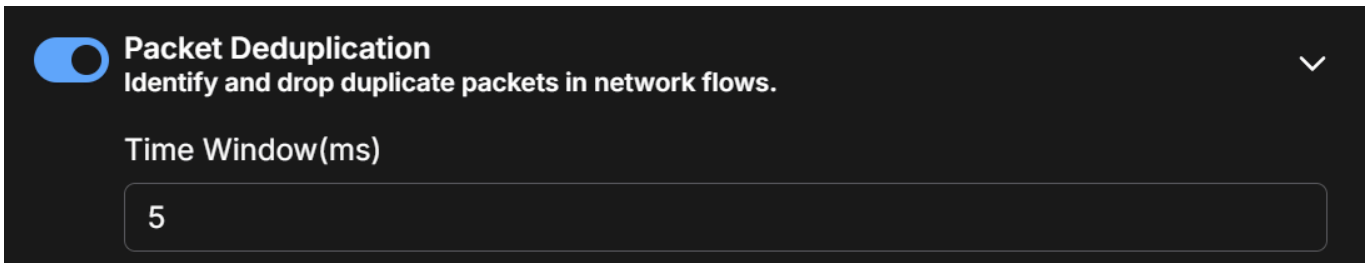


Enable this to build correlations between processes running on the sensor and host, and the IP address/port visible in traffic. The processes monitored include:

- log forwarder
- IDS
- maltrace
- aella_flow
- SSHD
- HTTP

You can set the sampling time interval.

Packet Deduplication



Enable this option for the sensor to perform packet deduplication. This reduces storage by removing duplicate data. However, it uses processing power on the sensor to analyze the data and can significantly slow system performance.

When this option is enabled, you also set the deduplication **Time Window**, which specifies in **ms** the time window within which the sensor considers an identical to be a duplicate and removed. When the sensor receives a packet, it checks whether an identical packet arrived within the configured time window. If a match is found, the later packet is discarded as a duplicate.

❗ Enable with caution due to the sensor resources required. If you have no need for deduplication, disable it.

How It Works

When a packet arrives from multiple SPAN ports, each copy travels a different path to the sensor and may arrive at slightly different times. If the gap between arrivals falls within the time window, the sensor recognizes them as duplicates and retains only the first.

Example 1 — Duplicate detected (Time Window = 5 ms)

```
Original Packet
  └─ SPAN port 1 → arrives at t = 0 ms

Same Packet
  └─ SPAN port 2 → arrives at t = 3 ms

Arrival gap: 3 ms < 5 ms → Deduplicated
```

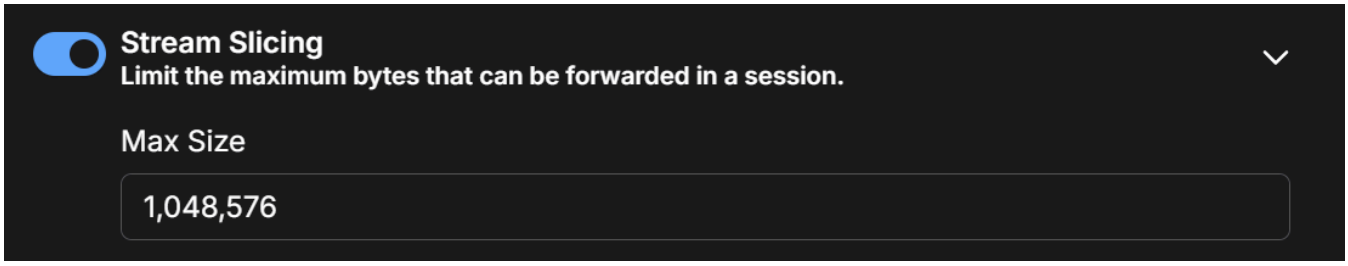
Example 2 — Duplicate not detected (Time Window = 5 ms)

```
Original Packet
  └─ SPAN port 1 → arrives at t = 0 ms

Same Packet
  └─ SPAN port 2 → arrives at t = 6 ms

Arrival gap: 6 ms > 5 ms → Not deduplicated
```

Stream Slicing



Enable this to perform TCP stream slicing, which truncates sessions at the specified length. This reduces the number of packets sent to the Data Processor.

Stream slicing is different from packet slicing – instead of truncating individual packets, stream slicing captures the first **N** bytes of a stream/session (the reassembled flow) and discards the rest. A stream is identified by all packets where the source/destination IP addresses, source/destination ports, and protocol match.

This has no effect if you use local file assembly, which requires all packets from a stream to perform reassembly.

- **Minimum:** 2024 bytes
- **Maximum:** 4,294,967,295 bytes

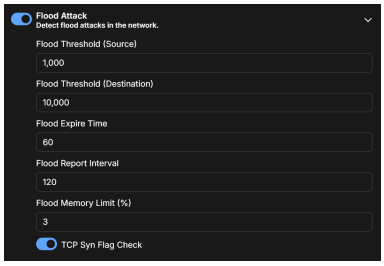
Handshake Failure



Enable this to detect handshake failures and allocate resources for that effort:

- **Time(s)** – amount of time the sensor collects failures to get to the threshold
- **Threshold** – the number of failures over the specified time before this detection is triggered
- **Report Interval(s)** – interval between reports to the DP if there are multiple failures
- **Memory Limit (%)** – maximum percentage of memory used by `aella_flow` for this detection

Flood Attack



Enable this to report flood attacks:

- **Flood Threshold (Source)** – the number of new session requests per second from a single source before an attack is reported
- **Flood Threshold (Destination)** – the number of new session requests per second from any source before an attack is reported
- **Flood Expire Time(s)** – the number of seconds without new session requests after which we stop considering it an attack, and the sensor sends a `session_end` message to the DP
- **Flood Report Interval(s)** – during an attack, the interval at which the sensor reports the attack to the DP (so that the DP isn't flooded with reports of the flood attack)
- **TCP Syn Flag Check** – only counts session requests if the SYN flag in the packet is set

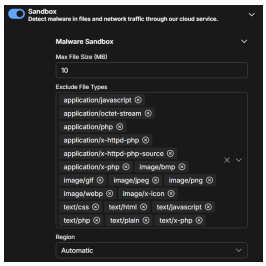
Buffering

If you enable buffering, the sensor buffers the data sent to Stellar Cyber. If the buffered data is received successfully, the sensor deletes the buffered data. If the buffer is full, the sensor stops buffering. The **Buffer Size** specifies the maximum amount of data buffered.

Sandbox

The malware sandbox feature enables the Stellar Cyber Platform to forward suspicious files extracted from network traffic, email attachments, or other sources to an external sandbox, which then executes files within a controlled virtual environment to determine if it exhibits malicious activity.

Enable the cloud sandbox to detect malware in network traffic. You must enable **Network Traffic** to enable the sandbox.



You can control the type and maximum size of scanned files, as well as the region of the sandbox used:

- **Max File Size (MB)** – The largest file that will be scanned. The default and maximum values are both 10 MB.
- **Exclude MIME Types** – Specifies which file types are scanned.

  Expand to see the file types that are excluded from scanning by default:

  - application/javascript
  - application/octet-stream
  - application/php
  - application/x-httpd-php
  - application/x-httpd-php-source
  - application/x-php
  - image/bmp
  - image/gif
  - image/jpeg
  - image/png
  - image/webp
  - image/x-icon
  - text/css
  - text/html
  - text/javascript
  - text/php
  - text/plain
  - text/x-php

  Expand to see the file types available for scanning inclusion/exclusion:

File Type	Extension
application/vnd.android.package- archive	*.apk*
application/x-apple-diskimage	*.img*, *.smi*, *.dmg*
data/email-rfc2822	*.eml*
application/java-archive	*.jar*
application/war-archive	*.war*
application/x-nuget	*.nupkg*

File Type	Extension
application/llappbundle-document	*.llappbundle*, *.llapp*, *.tar*
application/llappbundle-macos- executable	*.llappbundle*, *.llapp*, *.tar*
application/llappbundle-windows- executable	*.llappbundle*, *.llapp*, *.tar*
application/llappbundle-web-replay	*.llappbundle*, *.llapp*, *.tar*
application/x-xar	*.xar*, *.pkg*
application/zip	*.zip*
application/hangul-word	*.hwp*
application/x-chm	*.chm*
application/x-hwp	*.hwp*
application/msoffice-mime-xls	*.xls*
application/msoffice-mime-ppt	*.ppt*
application/msoffice-mime-doc	*.doc*
application/msoffice-xls	*.xls*
application/msoffice-xls-encrypted	*.xlsx*, *.xls*
application/msoffice-ppt-encrypted	*.pptx*, *.ppt*
application/msoffice-doc-encrypted	*.doc*, *.docx*
application/msoffice-ppt	*.pps*, *.ppt*
application/msoffice-doc	*.doc*
application/msoffice-publisher	*.pub*
application/vnd.oasis.opendocument	*.odp*, *.ods*, *.odt*, *.otg*,*.otp*, *.ott*, *.odg*
application/pdf	*.pdf*
application/wordperfect	*.wpd*
text/rtf	*.rtf*

File Type	Extension
application/x-shockwave-flash	*.swf*
application/x-spreadsheetml	*.xml*
application/x-wordprocessingml	*.xml*
application/vnd.adobe.xdp+xml	*.xdp*
application/msoffice-xlsx	*.xlsx*
application/msoffice-xlsm	*.xlsm*
application/msoffice-xlsb	*.xlsb*
application/msoffice-xltm	*.xltm*
application/msoffice-pptx	*.pptx*, *.ppsx*
application/msoffice-pptm	*.pptm*
application/msoffice-ppsx	*.ppsx*
application/msoffice-ppsm	*.ppsm*
application/msoffice-potm	*.potm*
application/msoffice-docx	*.docx*
application/msoffice-docm	*.docm*
application/msoffice-dotm	*.dotm*
application/vnd.ms-xpsdocument	*.xps*
application/oxps	*.oxps*
application/x-java-class	*.class*
application/x-com	*.com*
application/x-eicar	*.com*
application/x-dosexec	*.exe*
application/x-msi	*.msi*

File Type	Extension
application/x-ms-shortcut	*.url*, *.lnk*
application/x-mach-o-binary	*.bundle*, *.o*, *.dylib*
application/x-pe	*.sys*, *.exe*, *.dll*
application/x-rar-sfx-pe	*.exe*
application/x-zip-sfx-pe	*.exe*
application/x-7zip-sfx-pe	*.exe*
application/x-lastline-test	*.sys*, *.exe*, *.dll*
application/x-mach-o-fat-binary	*.bundle*, *.o*, *.dylib*
image/tiff	*.tiff*, *.tif*
image/svg	*.svg*
text/hta	*.hta*
text/vba	*.vba*
text/vbscript	*.vbs*
application/encodedvbscript	*.vbe*
text/x-msdos-batch	*.bat*, *.cmd*
application/javascript	*.js*
application/encodedjscript	*.jse*
text/x-powershell	*.psm1*, *.psd1*, *.ps1*
text/x-wsf	*.wsf*
text/html	*.htm*, *.html*

To exclude more MIME file types from scanning, click or tap in an empty space in the **Exclude MIME Types** field to reveal additional types and then select the ones you want to add to the exclusion list. When done, click or tap outside the field to exit the selection mode.

- **Region** – Choose the geographical region of the malware sandbox used by this profile. By default, this option is set to **Automatic** and the sensor chooses the sandbox with the

lowest latency relative to its own location. You can optionally override the **Automatic** option and choose a sandbox in a specific geographic location from the dropdown list.

Stellar Cyber sends files with suspected malware for analysis over HTTPS on TCP port 443 to sandboxes at the following URLs:

Region	URL
Australia	au.sandbox.stellarcyber.ai
Germany	de.sandbox.stellarcyber.ai
Japan	jp.sandbox.stellarcyber.ai
United Kingdom	uk.sandbox.stellarcyber.ai
United States of America	us.sandbox.stellarcyber.ai

If there is a perimeter firewall with strict outbound security policy rules, make sure it permits HTTPS on TCP 443 from the sensor to the URL for the chosen region. After the file is analyzed, the result is returned to the sensor, which then sends it to the Stellar Cyber Platform.

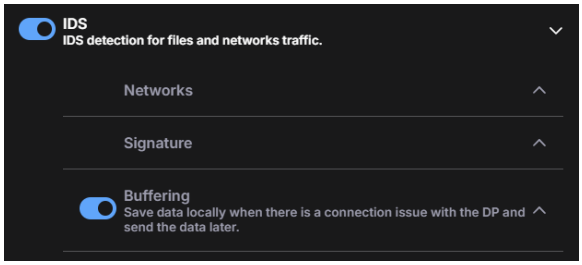
The Sandbox requires both Network Traffic and the Intrusion Detection System (IDS) to be enabled so it can function.

IDS is required to perform deep packet inspection (DPI) to analyze and reassemble data streams so that Stellar Cyber can extract files or payloads from network sessions and forward them to the malware sandbox. Also, the networks that you specify for IDS determine the scope of traffic that the sensor monitors for potential threats, including malware.

IDS

See the top of this page for recommended lessons about this section at Stellar Cyber Academy.

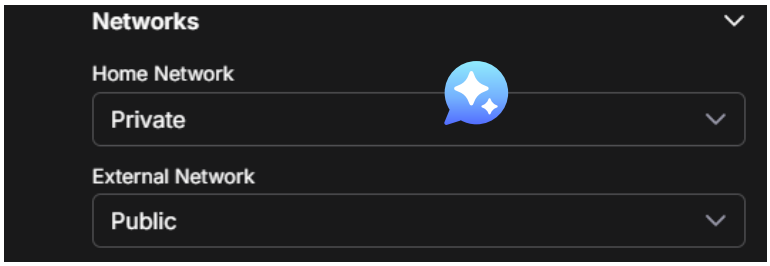
Enable IDS to detect malicious activity in network traffic. You must enable **Network Traffic** to enable IDS.



You can configure:

- Networks
- Signature
- Buffering

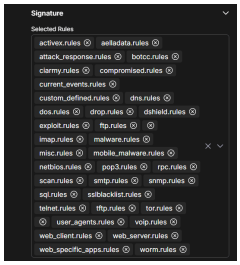
Networks



Choose the **Home Network**. If you choose **Specific**, you can enter the IP address range.

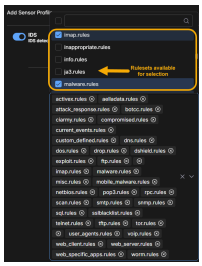
Choose the **External Network**. If you choose **Specific**, you can enter the IP address range.

Signature

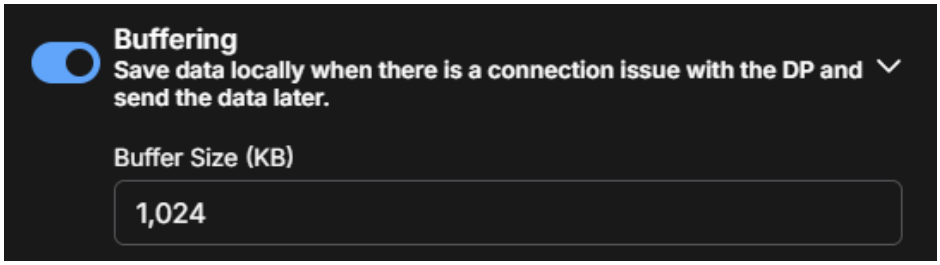


Choose the rule sets to use. The listed rule sets include both those integrated by Stellar Cyber from third-party threat intelligence and [the **custom_defined.rules** signature set created by importing your own security rules](#).

Stellar Cyber regularly pushes new rule sets so that they are available for selection in a sensor profile. Click in the **Selected Rules** list to see a dropdown of the rules available for selection. For example:



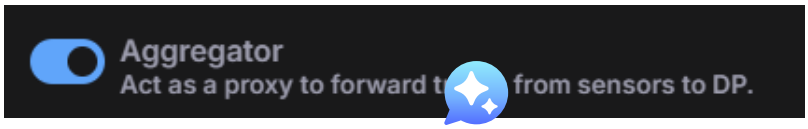
Buffering



If you enable buffering, the sensor buffers the data sent to Stellar Cyber. If the buffered data is received successfully, the sensor deletes the buffered data. If the buffer is full, the sensor stops buffering. The **Buffer Size** sets the maximum amount of data buffered.

Aggregator

See the [top of this page](#) for recommended lessons about this section at Stellar Cyber Academy.



Enable this for the sensor to act as a data aggregator for other sensors. The modular aggregator forwards aggregated traffic to the DP configured on the modular sensor. It also uses the receivers configured for the modular sensor, so you don't have to configure an aggregator profile.

As soon as you submit the modular sensor profile, you can configure sensors to send traffic to the modular aggregator.

Aggregator Rules

Keep in mind the following rules when using the **Aggregator** feature on the Modular Sensor:

- If you enable the **Aggregator** for a modular sensor, you cannot configure it to use another aggregator.
- If you configure your own proxy on the Modular Sensor, you cannot enable the **Aggregator** feature. The Modular Sensor does not support double-proxying (the aggregator is itself a form of proxy).
- TCP Port 8080 must be open between the sensor and the aggregator for successful communications.

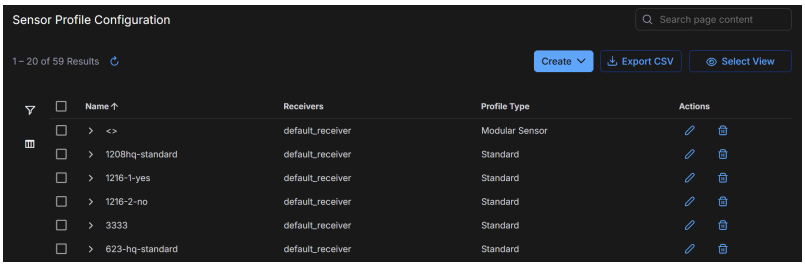
Tenable Nessus

See the top of this page for recommended lessons about this section at Stellar Cyber Academy.

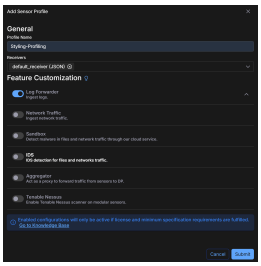
This feature is supported on modular sensors running v4.3.4 or later. Stellar Cyber recommends that you provision the modular sensor with an additional 1.5 GB of RAM and 1 vCPU when you enable this option.

Enable this feature to install a **tenable nessus** vulnerability scanner on your Stellar Cyber modular sensor, and link it to your nessus tenable.io account. After you have linked the scanner instance, you can initiate and monitor scans from the tenable.io

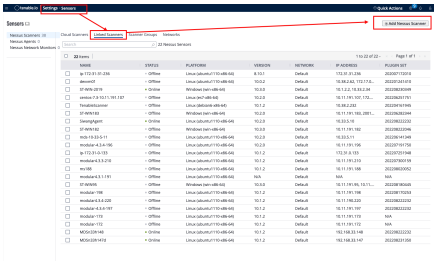
1. Click **System | DATA SOURCE MANAGEMENT | Sensors | Sensor Profiles**. The **Sensor Profile Configuration** page appears, with the **Sensor Profiles** tab displayed by default.



2. Click **Create** and select **Add Modular Sensor Profile**. The **ADD SENSOR PROFILE** screen appears.



3. Log into your **tenable.io** account as an administrative user (https://cloud.tenable.com).
4. Navigate to **Settings | Sensors**
5. From the **Nessus Scanners** tab, select the sub-tab for **Linked scanners**.
6. Click the link to **Add Nessus scanner.**

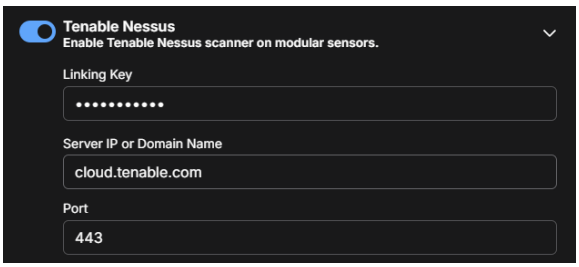


7. From the dialog that displays, copy the **Linking Key** and save it to a file.



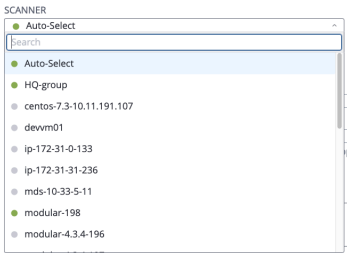
8. Return to the Stellar Cyber console, **Modular Sensor** profile you are adding or editing.
9. Locate the section labeled **Tenable Nessus**, and toggle the feature on.
10. In the panel that displays, enter the **Linking Key**.
11. Supply a value for the **Host**. By default, **Host** is set to cloud.tenable.com.
12. Supply a value for the **Port**. By default, **Port** is set to 443.

    A port must be specified for successful configuration.



13. Click **Submit**.

After this profile is associated with a sensor, you can select that sensor IP address when you create or edit a scan in the **Vulnerability Management | Scans** panel.



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